Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title:	Electric Networks
Course Number:	ELE 302
Semester/Year (e.g.F2016)	F2026

Instructor:	Dr. Andy Ye

Assignment/Lab Number:	2
Assignment/Lab Title:	The Step Response of Second-Order Circuits

Submission Date:	October 15, 2026
Due Date:	October 15, 2026

Student LAST Name	Student FIRST Name	Student Number	Section	Signature*
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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a “0” on the work, an “F” in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

- The frequency, **f** is equal to the number of repetitions per unit of time, and can be calculated from the period, **T**: $\mathbf{f}(\text{Hz}) = \frac{1}{\mathbf{T}(\text{sec})}$ or $\mathbf{T}(\text{sec}) = \frac{1}{\mathbf{f}(\text{Hz})}$.
  When frequency is given in radians/sec then the symbol, **ω** is used, where **ω** = 2π**f**
- Phase Shift, **Θ** of one sinusoidal signal with respect to another (of the same frequency) occurs when there is time-offset, **ΔT** between them. Phase-Shift => $\boldsymbol{\theta} = \frac{\Delta T}{T}.2\pi\ (\text{radians}) = \frac{\Delta T}{T}.360°(\text{degrees})$
- Sinusoidal AC voltage as a function of time: $v(t) = \mathbf{V_P}\cos(\boldsymbol{\omega}t + \boldsymbol{\theta}) = \mathbf{V_P}\cos(2\pi\mathbf{f}t + \boldsymbol{\theta})$

2.0 OBJECTIVES:

- To use Multisim circuit simulation software to plot the step response of second-order dynamic circuits.
- To use the oscilloscope to display the step response of second-order dynamic circuits.
- To measure the parameters that characterize the step response of second-order dynamic circuits.

3.0 REQUIRED LAB EQUIPMENT & PARTS:

- Function Generator (FG) and Oscilloscope.
- ELE202 Lab Kit and ELE302 Lab Kit: various components, breadboard, wires and jumpers.

4.0 PRE-LAB ASSIGNMENT (3 marks with 0.75 marks for each step):

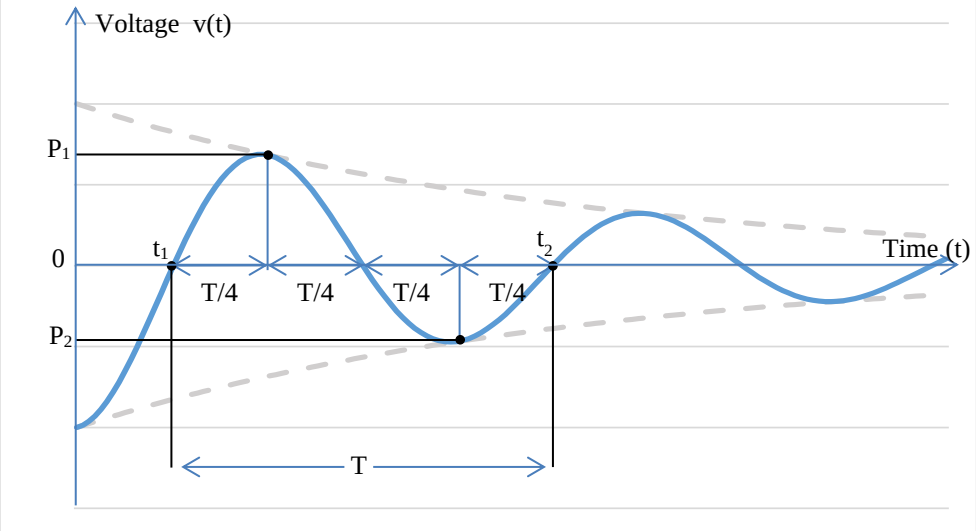


Figure 2.0: An Exponentially Decaying Sinusoidal Function

(a) Step 1: The circuit shown in **Figure 2.0** is an exponentially decaying sinusoidal function that can be characterized by the function: $v(t) = Ae^{-\sigma t}\cos(\boldsymbol{\omega}t + \boldsymbol{\theta})$. The figure defines three variables T, P_1, and P_2. t_1 is the time of the first zero-crossing of v(t) and t_2 is time for the third zero-crossing of v(t). T is the time difference between t_1 and t_2. P_1 is the value of v(t) one quarter of the way between the first zero-crossing and the third zero-crossing (i.e. $P_1 = v(t_1 + \frac{T}{4})$). Finally, P_2 is the value of v(t) three quarters of the way between the first zero-crossing and the third zero-crossing (i.e. $P_2 = v(t_1 + 3\frac{T}{4})$).

i) Find an expression for ω in terms of T.

Pre-Lab workspace (show your analysis here)

Since t_3 is the third zero crossing while t_1 is the first, t_1 and t_3 are one period apart. That means T is exactly 1 period of the cosine function. Since $T = 2\pi w$, $\omega = \frac{2\pi}{T}$

ii) Find an expression for σ in terms of P_1, P_2, and T.

Pre-Lab workspace (show your analysis here)

$P_1 = Ae^{-\sigma(t_1 + T/4)}(1)$ → as the function peaks

$P_2 = Ae^{-\sigma(t_1 + 3/4T)}(-1)$ → as function troughs

$$\frac{P_1}{P_2} = \frac{Ae^{-\sigma(t_1+T/4)}}{-Ae^{-\sigma(t_1+3T/4)}}$$

$$\frac{P_1}{P_2} = -e^{-\sigma[t_1+T/4-(t_1+3T/4)]}$$

$$\frac{P_1}{P_2} = -e^{\sigma(T/2)}$$

$$\sigma\left(\frac{T}{2}\right) = \ln\left(\frac{-P_1}{P_2}\right)$$

$$\sigma = \frac{2}{T}\ln\left|\frac{P_1}{P_2}\right|$$

iii) Find an expression of θ in terms of t_1 and T.

Pre-Lab workspace (show your analysis here)

@ t_1, cosine is 0 while increasing. ∴ the terms inside the cosine is equivilent to $-\frac{\pi}{2}$.

$$\omega t_1 + \theta = -\frac{\pi}{2}$$

$$\theta = -\frac{\pi}{2} - \omega t_1$$

$$= -\frac{\pi}{2} - \frac{2\pi}{T}t_1$$

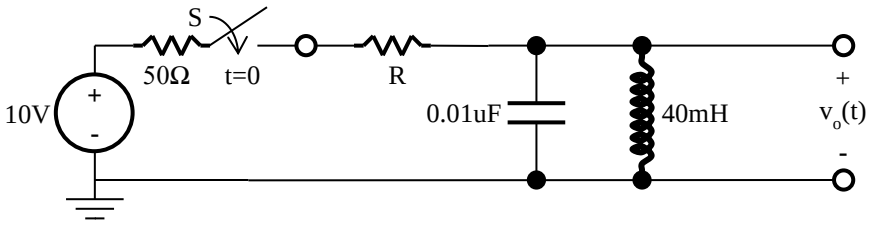
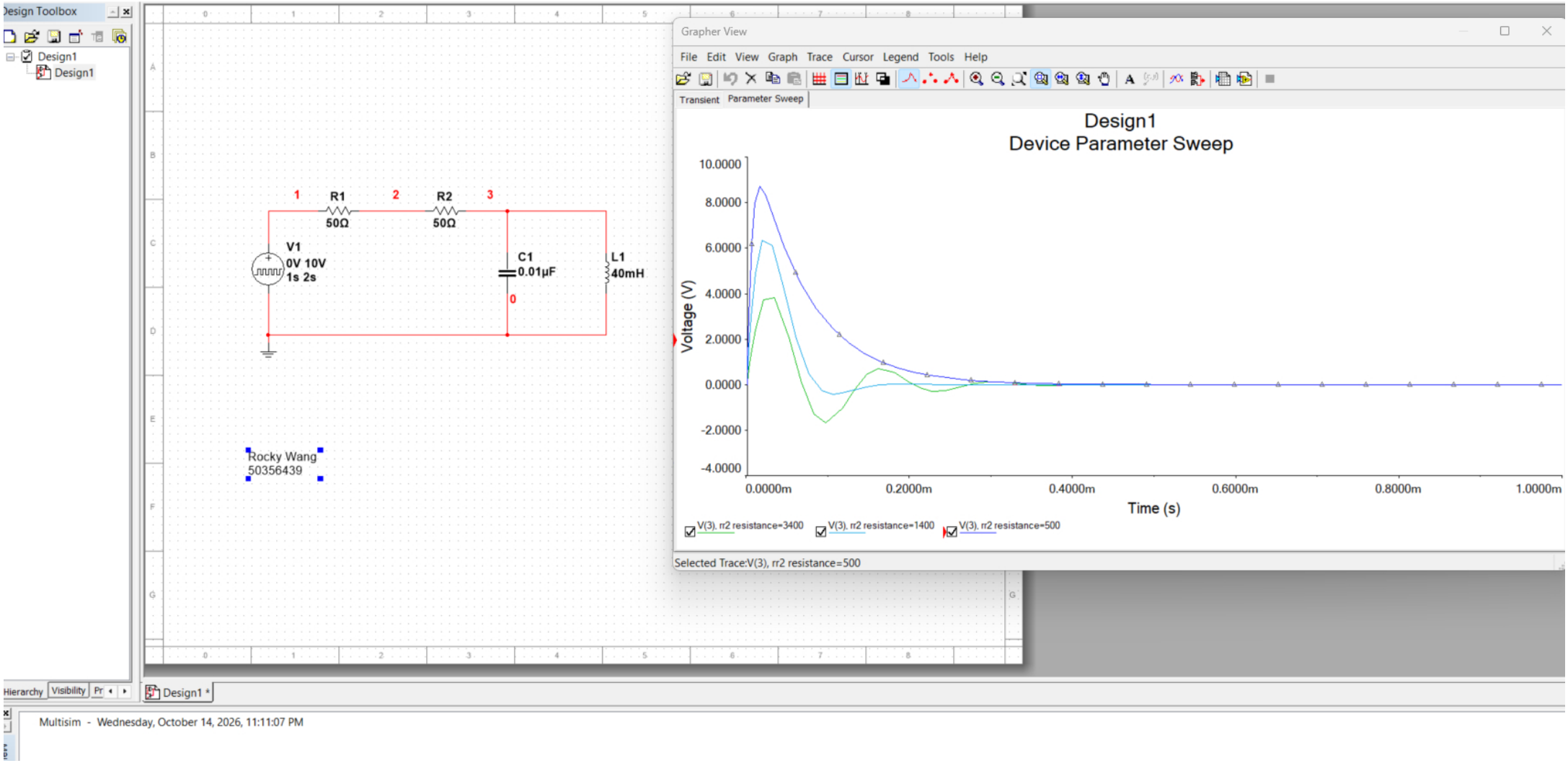


Figure 2.0: Second-Order Bandpass Circuit

(b) Step 2: Consider the dynamic circuit shown in **Figure 2.0**. The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$] when R=1.4kΩ, 3.4kΩ, and 500Ω.

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
1
R1
50Ω
2
R2
50Ω
3
V1
0V 10V
1s 2s
C1
0.01µF
L1
40mH
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient
Parameter Sweep
Design1
Device Parameter Sweep
Voltage (V)
10.0000
8.0000
6.0000
4.0000
2.0000
0.0000
-2.0000
-4.0000
0.0000m
0.2000m
0.4000m
0.6000m
0.8000m
1.0000m
Time (s)
V(3), rr2 resistance=3400
V(3), rr2 resistance=1400
V(3), rr2 resistance=500
Selected Trace:V(3), rr2 resistance=500
Hierarchy
Visibility
Design1 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM

ii) Use the plots to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = A\boldsymbol{e}^{-\sigma t}\sin(\omega t)$ for R=3.4kΩ.

Pre-Lab workspace (show your analysis here)

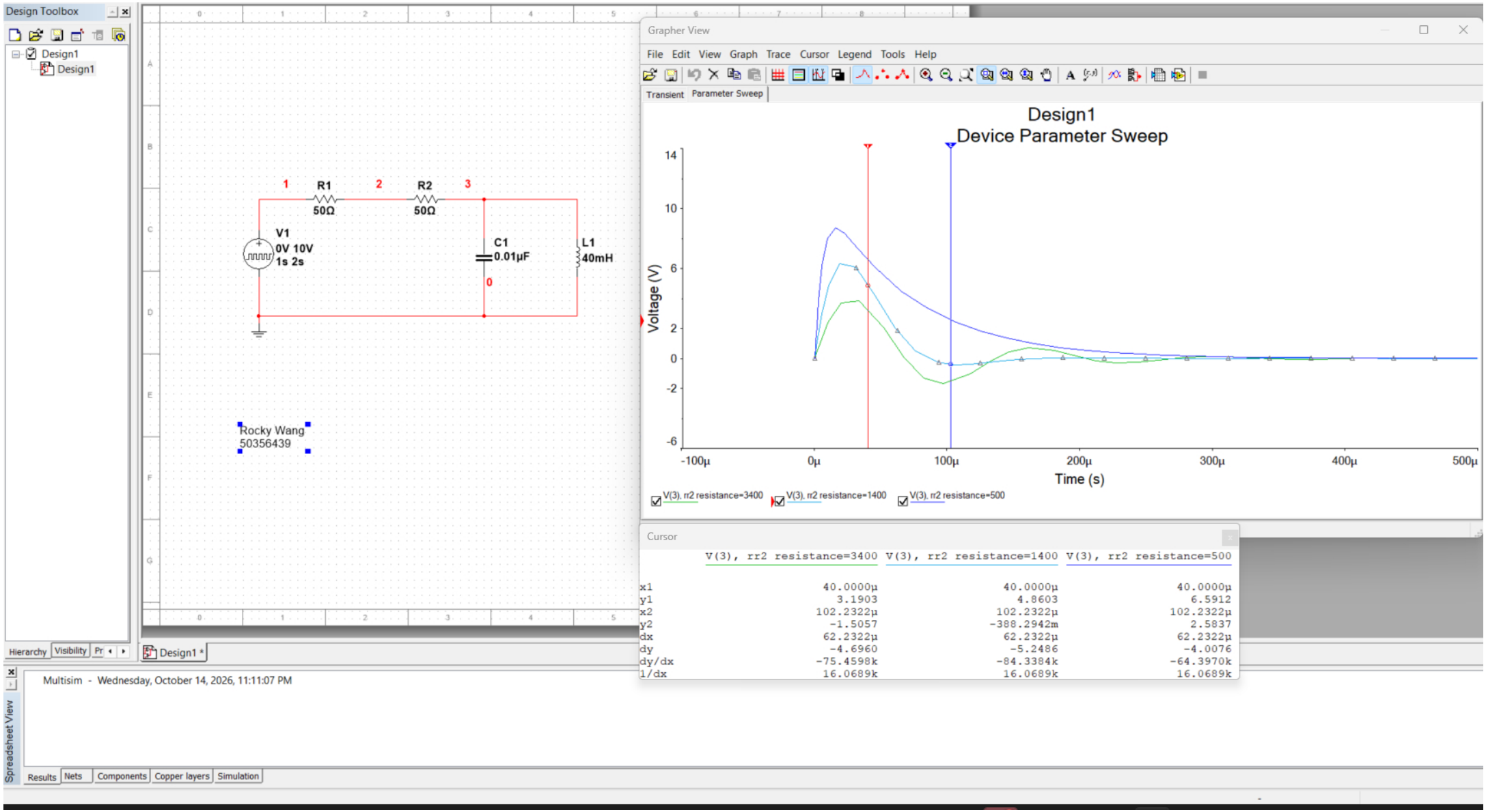
$\Delta x = 136.3\,\mu s = T$ | $t_1 = 0s$ | $P_1 = 3.77\,V$

$T/4 = 34.0774\,\mu s$ | $t_1 + T/4 = 34.0774\,\mu s$ | $P_2 = -1.5057\,V$

$\frac{3}{4}T = 102.2322\,\mu s$ | $t_1 + \frac{3}{4}T = 102.2322\,\mu s$

$$\sigma = \left(\frac{2}{136.3\mu}\right)\ln\left(\frac{3.77}{1.5057}\right) = 13470/s$$

$$\omega = \frac{2\pi}{136.3\mu} = 46100\ rad/s$$


Design Toolbox
Design1
Design1
1
R1
50Ω
2
R2
50Ω
3
V1
0V 10V
1s 2s
C1
0.01µF
0
L1
40mH
Rocky Wang
50356439
Design1 *
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient Parameter Sweep
Design1
Device Parameter Sweep
Voltage (V)
Time (s)
V(3), rr2 resistance=3400
V(3), rr2 resistance=1400
V(3), rr2 resistance=500
Cursor
V(3), rr2 resistance=3400 V(3), rr2 resistance=1400 V(3), rr2 resistance=500
x1 40.0000µ 40.0000µ 40.0000µ
y1 3.1903 4.8603 6.5912
x2 102.2322µ 102.2322µ 102.2322µ
y2 -1.5057 -388.2942m 2.5837
dx 62.2322µ 62.2322µ 62.2322µ
dy -4.6960 -5.2486 -4.0076
dy/dx -75.4598k -84.3384k -64.3970k
1/dx 16.0689k 16.0689k 16.0689k
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Spreadsheet View
Results Nets Components Copper layers Simulation

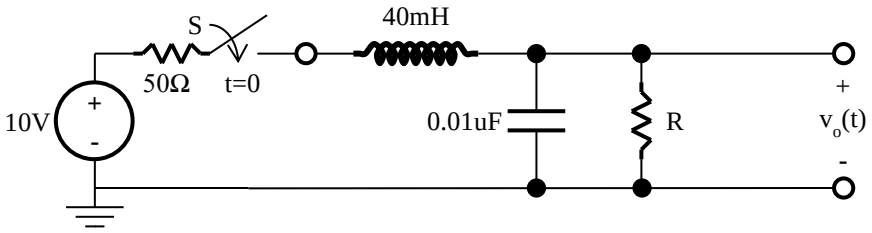
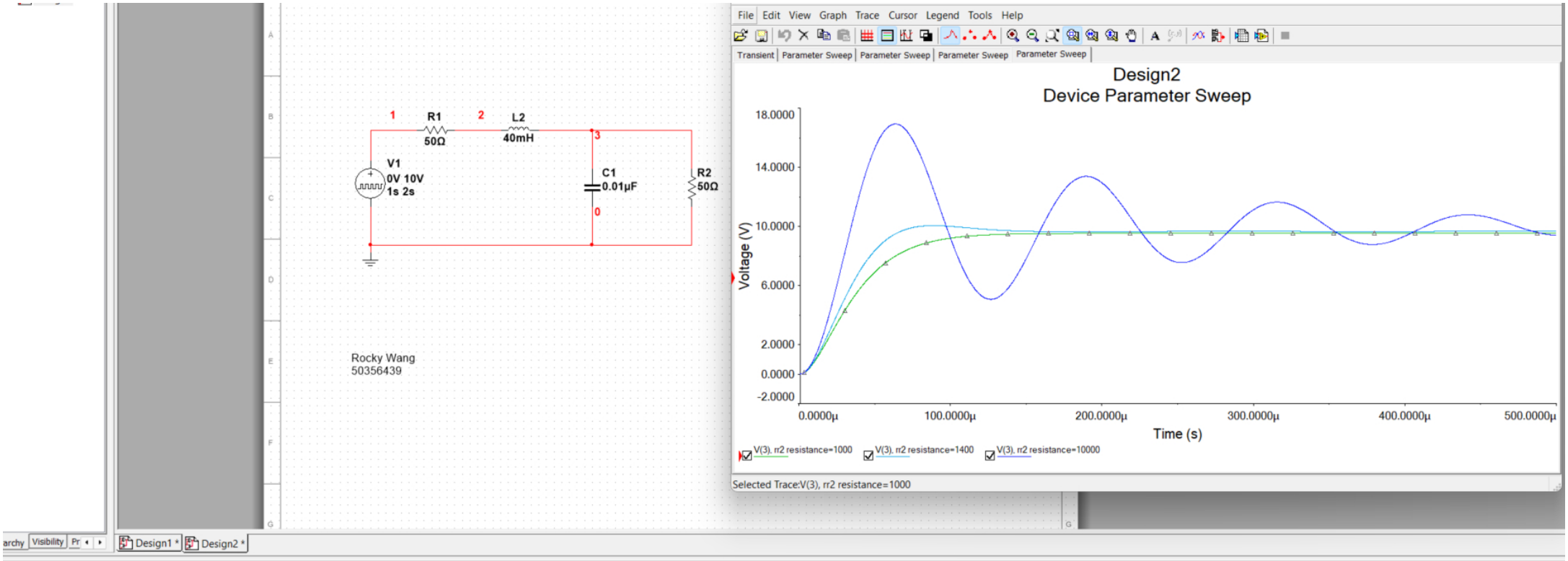


Figure 3.0: Second-Order Lowpass Circuit

(c) Step 3: Consider the dynamic circuit shown in **Figure 3.0.** The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$] when R=1kΩ, 1.4kΩ, and 10kΩ.

Pre-Lab workspace (show your analysis here)



Multisim - Wednesday, October 14, 2026, 11:11:07 PM

ii) Use the plots to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = B + Ae^{-\sigma t}\cos(\omega t + \theta)$ for R=10kΩ.

Pre-Lab workspace (show your analysis here)

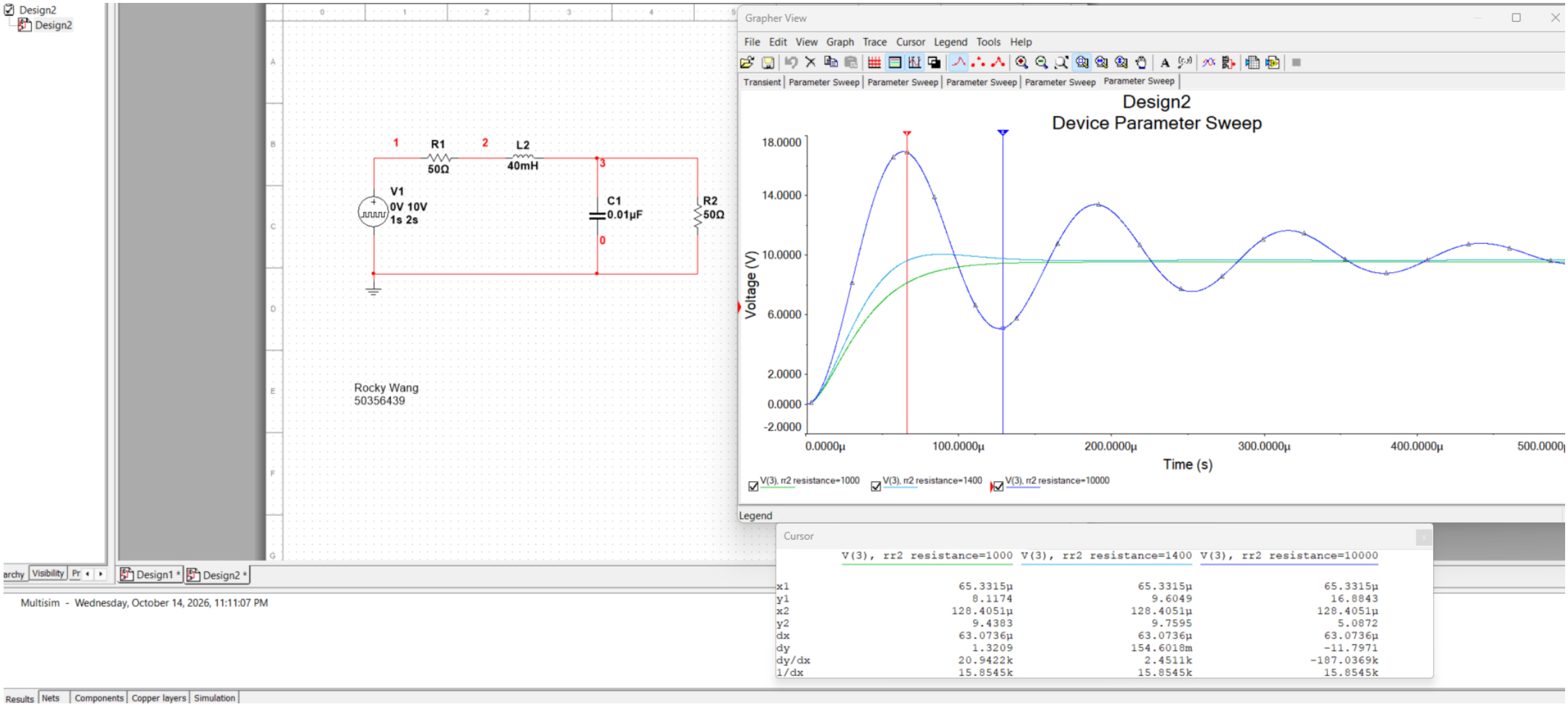
$\Delta x = 126.1472\,\mu s = T$ | $t_1 = 33.7947\,\mu s$ | $P_1 = 16.8843 - 9.95 = 6.934\,V$

$T/4 = 31.5368\,\mu s$ | $t_1 + T/4 = 65.3315\,\mu s$ | $P_2 = 5.0872 - 9.95 = -4.863\,V$

$\frac{3}{4}T = 94.6104\,\mu s$ | $t_1 + \frac{3}{4}T = 128.4051\,\mu s$

$$\sigma = \left(\frac{2}{126.1472\mu}\right)\ln\left(\frac{6.934}{4.863}\right) = 5626/s$$

$$\omega = \frac{2\pi}{126.1472\mu} = 49810\ rad/s$$


Design2
Design2
1
R1
50Ω
2
L2
40mH
3
V1
0V 10V
1s 2s
C1
0.01µF
R2
50Ω
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep
Design2
Device Parameter Sweep
Voltage (V)
18.0000
14.0000
10.0000
6.0000
2.0000
0.0000
-2.0000
0.0000µ
100.0000µ
200.0000µ
300.0000µ
400.0000µ
500.0000
Time (s)
V(3), rr2 resistance=1000
V(3), rr2 resistance=1400
V(3), rr2 resistance=10000
Legend
Cursor
V(3), rr2 resistance=1000 V(3), rr2 resistance=1400 V(3), rr2 resistance=10000
x1 65.3315µ 65.3315µ 65.3315µ
y1 8.1174 9.6049 16.8843
x2 128.4051µ 128.4051µ 128.4051µ
y2 9.4383 9.7595 5.0872
dx 63.0736µ 63.0736µ 63.0736µ
dy 1.3209 154.6018m -11.7971
dy/dx 20.9422k 2.4511k -187.0369k
1/dx 15.8545k 15.8545k 15.8545k
Visibility
Design1 *
Design2 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Results
Nets
Components
Copper layers
Simulation

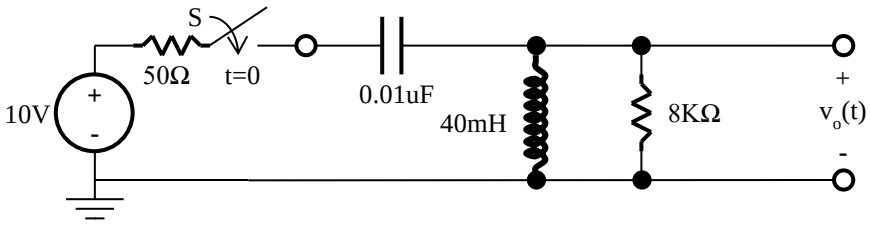
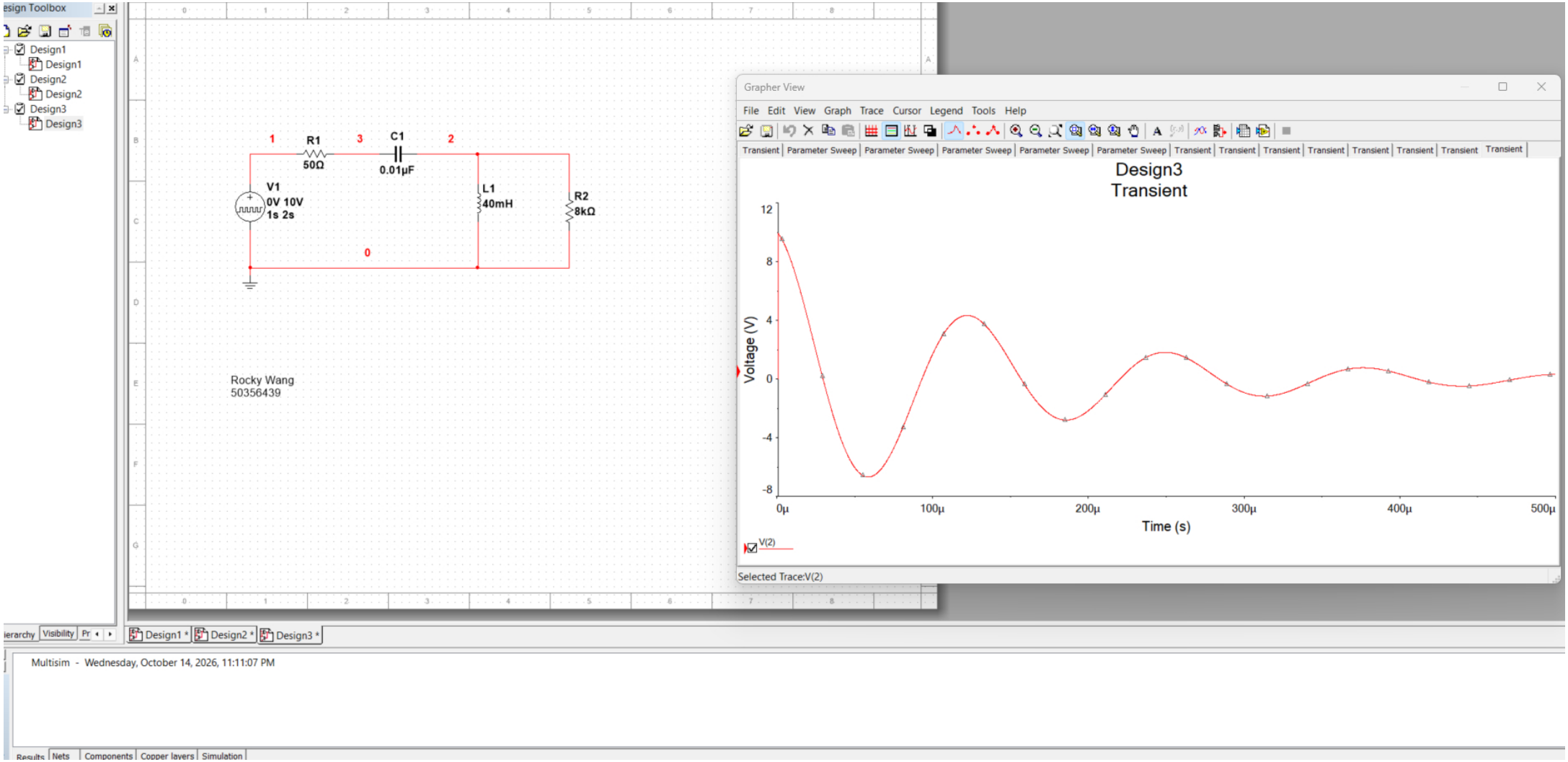


Figure 4.0: Second-Order Highpass Circuit

(d) Step 4: Consider the dynamic circuit shown in **Figure 4.0.** The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$].

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
Design2
Design2
Design3
Design3
1
R1
50Ω
3
C1
0.01µF
2
V1
0V 10V
1s 2s
L1
40mH
R2
8kΩ
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Design3
Transient
Voltage (V)
Time (s)
12
8
4
0
-4
-8
0µ
100µ
200µ
300µ
400µ
500µ
V(2)
Selected Trace:V(2)
Design1 *
Design2 *
Design3 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Results
Nets
Components
Copper layers
Simulation

ii) Use the plot to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = Ae^{-\sigma t}\cos(\omega t + \theta)$.

Pre-Lab workspace (show your analysis here)

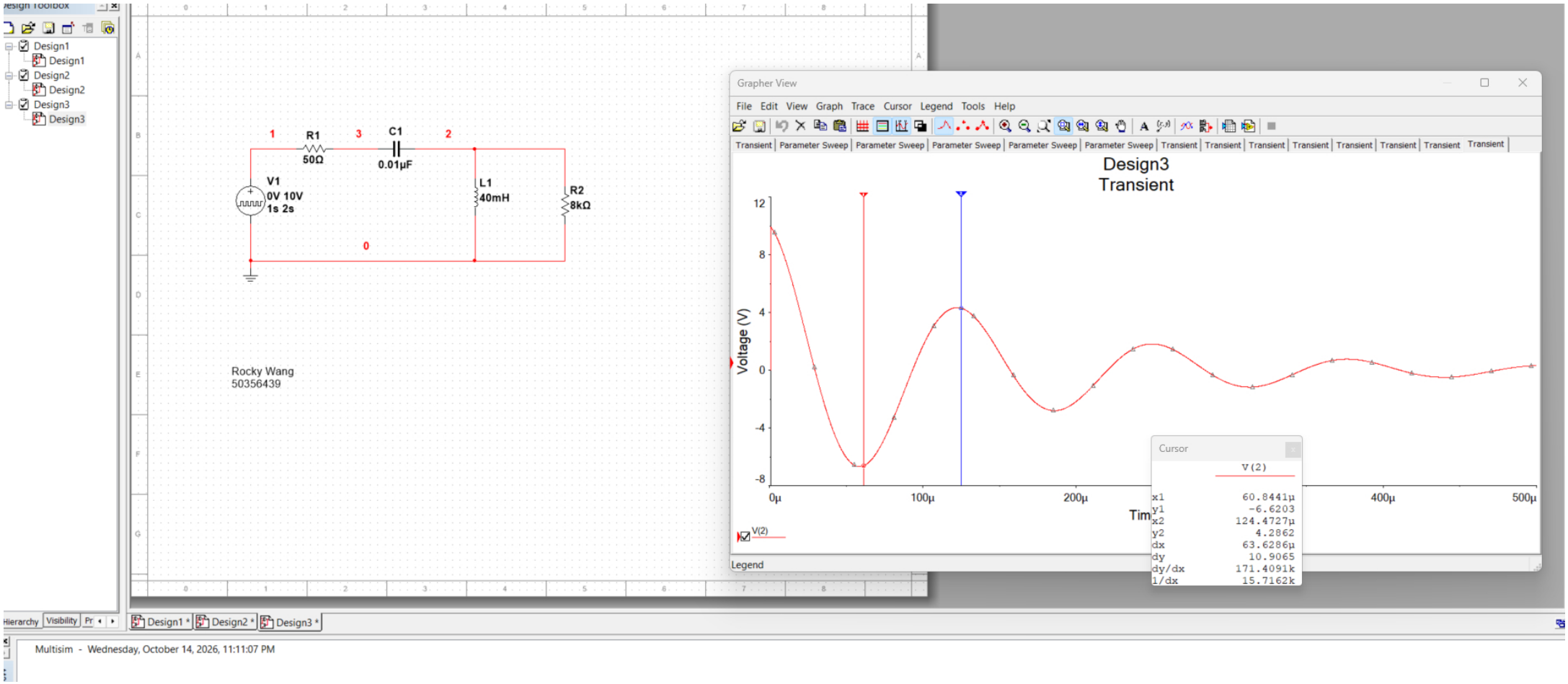
$\Delta x = 127.2572\,\mu s$ | $t_1 = 29.0298\,\mu s$ | $P_1 = -6.6203\,V$

$T/4 = 31.8143\,\mu s$ | $t_1 + T/4 = 60.8441\,\mu s$ | $P_2 = 4.2862\,V$

$3/4\,T = 95.4429\,\mu s$ | $t_1 + 3/4\,T = 124.4727\,\mu s$

$$\sigma = \left(\frac{2}{127.2572\mu}\right)\ln\left(\frac{6.6203}{4.2862}\right) = 6832/s$$

$$\omega = \frac{2\pi}{127.2572\mu} = 49370\ rad/s$$


Design Toolbox
Design1
Design1
Design2
Design2
Design3
Design3
1
R1
50Ω
3
C1
0.01µF
2
V1
0V 10V
1s 2s
L1
40mH
R2
8kΩ
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Transient | Transient | Transient | Transient | Transient | Transient | Transient | Transient
Design3
Transient
Voltage (V)
12
8
4
0
-4
-8
0µ
100µ
200µ
400µ
500µ
V(2)
Legend
Cursor
V(2)
x1 60.8441µ
y1 -6.6203
x2 124.4727µ
y2 4.2862
dx 63.6286µ
dy 10.9065
dy/dx 171.4091k
1/dx 15.7162k
Hierarchy
Visibility
Design1 *
Design2 *
Design3 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM